

Discrete Global Grid Thermodynamic Initialization: Conserved Multiphase Baselines for Planetary Earth System Monads

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Abstract

Discrete Global Grid Systems (DGGS) using hexagonal hierarchical tessellations, such as the Uber H3 framework, offer equal-area, low-distortion planetary spatial discretizations. In computational Earth system simulations and non-equilibrium ecological models, spatial cells must not serve as mere geometric primitives; they must represent thermodynamically closed or flux-coupled multiphase physical control volumes. Initializing discrete grid cells without rigorous thermodynamic constraints frequently causes numerical energy drift, non-physical entropy drops, and mass inconsistencies during cross-scale aggregation or advective-diffusive transport.

In this paper, we formalize the baseline thermodynamic initialization of H3 discrete spatial cells at Standard Temperature and Pressure (STP; $T_0 = 288.15$ K, $P_0 = 101,325$ Pa). We establish exact area scaling across hierarchical resolutions $r \in [0, 15]$ derived from icosahedral geodesics, hydrostatic column air mass conservation, August-Roche-Magnus psychrometric partitioning, and active lithospheric-biospheric stoichiometric inventories. Furthermore, we cast the resulting tensor factory within a functional monadic architecture, wherein the baseline tensor acts as the unit morphism η for a spatial monad $\mathcal{M}(\text{H3CellThermodynamicState})$. This formulation guarantees strict compliance with the First and Second Laws of Thermodynamics, establishing a stable, reproducible foundation for multi-scale planetary simulation engines.

Keywords: Discrete Global Grid Systems, Hexagonal Hierarchies, Non-Equilibrium Thermodynamics, Earth System Modeling, Monadic State Transitions, Mass-Energy Conservation

1 Introduction

The representation of continuous planetary biophysical dynamics on discrete computational substrates presents deep architectural and physical challenges. Traditional latitude-longitude raster grids introduce polar singularities and severe area distortions, confounding non-equilibrium transport equations and spatial mass fluxes [1]. Discrete Global Grid Systems (DGGS), particularly hexagonal aperture systems based on icosahedral projections such as Uber’s H3 [2], resolve geometric uniformity constraints by providing equal-area topological neighborhood relations and isotropic diffusion operators.

However, computational ecology and Earth system models frequently treat spatial cells as arbitrary key-value stores or unconstrained parametric state vectors. In the absence of an axiomatic baseline, dynamically generated or subdivided cells suffer from thermodynamic drift:

1. **Mass Inconsistency:** Arbitrary assignment of column atmospheric pressures or chemical species fractions without explicit area scaling introduces phantom mass creation or destruction ex nihilo.

2. **Thermodynamic Discontinuity:** Instantiating temperature or enthalpy outside of local thermodynamic equilibrium (LTE) leads to non-physical negative entropy generation ($\sigma < 0$) upon numerical initialization.
3. **Coupling Fragility:** Downstream trophic, geochemical, and hydrologic operators require deterministic, immutable baseline stocks across all spherical envelopes—atmosphere, hydrosphere, lithosphere, and biosphere.

To resolve these challenges within the open-source planetary simulation architecture *Web of Life* (<https://github.com/pascalranoroarijaona/WebOfLife>), Sprint 044 introduces the baseline state tensor factory `createDefaultH3CellThermodynamicState`. This paper provides the rigorous mathematical, stoichiometric, and category-theoretic formulation governing this factory.

2 Geodesic Area Scaling & Discrete Global Grid Geometry

The H3 DGGS partitions the spherical Earth ($R_{\oplus} = 6.3710088 \times 10^6$ m, total surface area $A_{\oplus} = 4\pi R_{\oplus}^2 \approx 5.100656 \times 10^{14}$ m²) by projecting a regular icosahedron onto the sphere, yielding 122 base cells at resolution $r = 0$ (comprising 110 hexagons and 12 pentagons).

Accounting for the spherical deficit of the 12 pentagons (5/6 the area of a hexagon), the nominal base hexagonal area $\bar{A}_0$ is given by:

$$\bar{A}_0 = \frac{A_{\oplus}}{110 + 12 \left(\frac{5}{6}\right)} = \frac{A_{\oplus}}{120} \approx 4.357419 \times 10^{12} \text{ m}^2 \quad (1)$$

For hierarchical apertures of factor 7, the area $A(r)$ of an H3 cell at resolution $r \in [0, 15]$ scales strictly as:

$$A(r) = \bar{A}_0 \cdot 7^{-r} \quad (2)$$

Table 1: Geodesic surface area scaling across select H3 hierarchical resolutions.

Resolution (r)	Cell Area ($A(r)$, m ²)	Equivalent Area (km ²)	Characteristic Diameter (m)
0	4.3574×10^{12}	4,357,419	2,243,628
2	8.8927×10^{10}	88,927	320,518
4	1.8148×10^9	1,815	45,788
6	3.7037×10^7	37.04	6,541
8	7.5587×10^5	0.756	934.5
10	1.5426×10^4	0.0154	133.5
12	3.1481×10^2	0.000315	19.07
15	9.1782×10^{-01}	9.18×10^{-7}	1.029

Every thermodynamic state tensor is bound to its geometric footprint $A(r)$, ensuring that all intensive specific quantities scale linearly to absolute extensive stocks.

3 Physical Formulation of the Baseline State Tensor

The reference state is constructed at Standard Temperature and Pressure (STP):

$$T_0 = 288.15 \text{ K} \quad (15.00^\circ \text{C}) \quad (3)$$

$$P_0 = 101,325.0 \text{ Pa} \quad (1013.25 \text{ hPa}) \quad (4)$$

$$g_0 = 9.80665 \text{ m} \cdot \text{s}^{-2} \quad (5)$$

3.1 Hydrostatic Column Mass and Atmospheric Stoichiometry

Under the hydrostatic assumption ($\partial P / \partial z = -\rho g_0$), the vertical integration of the atmospheric density column yields the exact column mass density:

$$\sigma_{\text{atm}} = \int_0^\infty \rho(z) dz = \frac{P_0}{g_0} \approx 10,332.2745 \text{ kg} \cdot \text{m}^{-2} \quad (6)$$

The total atmospheric gas mass enclosed by an H3 cell of area A is therefore:

$$M_{\text{atm}}(A) = A \cdot \frac{P_0}{g_0} \quad (7)$$

Atmospheric moisture is governed by the August-Roche-Magnus formulation for saturation vapor pressure $e^*(T)$:

$$e^*(T) = 610.94 \cdot \exp\left(\frac{17.625 \cdot (T - 273.15)}{T - 273.15 + 243.04}\right) \text{ Pa} \quad (8)$$

At $T_0 = 288.15 \text{ K}$, $e^*(288.15) \approx 1705.62 \text{ Pa}$. Given a standard mean ambient relative humidity $\phi = 0.60$, the partial pressure of water vapor is:

$$P_{\text{H}_2\text{O}} = \phi \cdot e^*(T_0) \approx 1023.37 \text{ Pa} \quad (9)$$

The dry atmospheric partial pressure is $P_{\text{dry}} = P_0 - P_{\text{H}_2\text{O}} = 100,301.63 \text{ Pa}$.

The resulting wet-air mole fraction for vapor is $x_{\text{H}_2\text{O}} = P_{\text{H}_2\text{O}}/P_0 \approx 0.01009987$. The dry molar constituents ($\text{N}_2 : 0.780840$, $\text{O}_2 : 0.209460$, $\text{CO}_2 : 0.000420$, Trace: 0.009280) are scaled by $(1 - x_{\text{H}_2\text{O}})$:

$$x_{\text{N}_2} = 0.780840 \cdot (1 - x_{\text{H}_2\text{O}}) \approx 0.772953 \quad (10)$$

$$x_{\text{O}_2} = 0.209460 \cdot (1 - x_{\text{H}_2\text{O}}) \approx 0.207344 \quad (11)$$

$$x_{\text{CO}_2} = 0.000420 \cdot (1 - x_{\text{H}_2\text{O}}) \approx 0.00041576 \quad (12)$$

$$x_{\text{trace}} = 0.009280 \cdot (1 - x_{\text{H}_2\text{O}}) \approx 0.00918627 \quad (13)$$

The effective molar mass of the humid atmospheric column is:

$$\bar{M}_{\text{atm}} = \sum_k x_k M_k \approx 0.028854 \text{ kg} \cdot \text{mol}^{-1} \quad (14)$$

where $M_{\text{N}_2} = 0.0280134$, $M_{\text{O}_2} = 0.0319988$, $M_{\text{CO}_2} = 0.0440095$, $M_{\text{H}_2\text{O}} = 0.0180153$, and $M_{\text{trace}} = 0.0399480 \text{ kg} \cdot \text{mol}^{-1}$.

Total moles of atmospheric column gas per cell:

$$N_{\text{atm}}(A) = \frac{M_{\text{atm}}(A)}{\bar{M}_{\text{atm}}} \quad (15)$$

3.2 Multiphase Planetary Envelopes

Hydrosphere Stock: Standard terrestrial baseline initializes a default surface liquid retention depth of 50 mm ($\sigma_{\text{liquid}} = 50.0 \text{ kg} \cdot \text{m}^{-2}$), zero cryospheric ice ($\sigma_{\text{ice}} = 0$), and continental salinity ($S_0 = 0 \text{ PSU}$):

$$M_{\text{hydro}}(A) = 50.0 \cdot A \quad (16)$$

Lithosphere Active Soil Column: For an upper pedospheric active layer of depth $z = 1.0 \text{ m}$ and dry bulk density $\rho_{\text{bulk}} = 1300.0 \text{ kg} \cdot \text{m}^{-3}$, stocks partition into:

$$M_{\text{SOC}}(A) = 12.0 \cdot A \quad (\text{Soil Organic Carbon}) \quad (17)$$

$$M_{\text{mineral}}(A) = (1300.0 - 12.0) \cdot A = 1288.0 \cdot A \quad (18)$$

$$M_{\text{soil_water}}(A) = 200.0 \cdot A \quad (\theta_v = 0.20 \text{ m}^3/\text{m}^3) \quad (19)$$

Biosphere Baseline Stocks: Representative global terrestrial biomass densities:

$$M_{\text{auto}}(A) = 2.50 \cdot A \quad (\text{Autotroph primary producers}) \quad (20)$$

$$M_{\text{hetero}}(A) = 0.015 \cdot A \quad (\text{Heterotrophic consumers}) \quad (21)$$

$$M_{\text{detritus}}(A) = 0.75 \cdot A \quad (\text{Necromass / litter layer}) \quad (22)$$

4 Energy and Entropy Formulations

4.1 First Law Formulation (Internal Energy)

Sensible heat is integrated from absolute zero (0 K). Atmospheric constant-volume heat capacity for the mixture is:

$$c_{v,\text{atm}} = c_{p,\text{atm}} - \frac{R}{M_{\text{atm}}} = 1005.0 - \frac{8.314463}{0.028854} \approx 716.85 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1} \quad (23)$$

Latent heat of vaporization at temperature T_0 :

$$L_v(T_0) = 2.501 \times 10^6 - 2370.0 \cdot (T_0 - 273.15) \approx 2.46545 \times 10^6 \text{ J} \cdot \text{kg}^{-1} \quad (24)$$

The total internal energy U_{cell} is the additive sum of atmospheric sensible and latent energy, hydrospheric energy, lithospheric energy, and biospheric chemical/sensible heat:

$$U_{\text{cell}} = U_{\text{atm}} + U_{\text{hydro}} + U_{\text{litho}} + U_{\text{bio}} \quad (25)$$

where:

$$U_{\text{atm}} = M_{\text{atm}} c_{v,\text{atm}} T_0 + (n_{\text{H}_2\text{O}} M_{\text{H}_2\text{O}}) L_v(T_0) \quad (26)$$

$$U_{\text{hydro}} = M_{\text{hydro}} c_{v,\text{liquid}} T_0 \quad (27)$$

$$U_{\text{litho}} = (M_{\text{mineral}} c_{\text{mineral}} + M_{\text{SOC}} c_{\text{SOC}} + M_{\text{soil_water}} c_{v,\text{liquid}}) T_0 \quad (28)$$

$$U_{\text{bio}} = ((M_{\text{auto}} + M_{\text{hetero}}) c_{\text{bio}} + M_{\text{detritus}} c_{\text{detritus}}) T_0 \quad (29)$$

Heat capacities utilized: $c_{v,\text{liquid}} = 4184.0$, $c_{\text{mineral}} = 830.0$, $c_{\text{SOC}} = 1800.0$, $c_{\text{bio}} = 3600.0$, and $c_{\text{detritus}} = 1500.0 \text{ J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$.

4.2 Second Law Formulation (Equilibrium Reference Entropy)

The absolute baseline entropy S_{cell} represents local thermodynamic equilibrium at STP. For atmospheric ideal gas species i :

$$S_i = n_i \left[S_{i,298.15}^\circ + c_{p,i} \ln \left(\frac{T_0}{298.15} \right) - R \ln \left(\frac{x_i P_0}{P^\circ} \right) \right] \quad (30)$$

with $P^\circ = 10^5 \text{ Pa}$, and S° representing standard molar entropies (N_2 : 191.61, O_2 : 205.15, CO_2 : 213.78, $\text{H}_2\text{O}(g)$: 188.84 $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

Condensed phases contribute specific empirical entropies:

$$S_{\text{condensed}} = M_{\text{liquid}} s_{\text{liquid}}^\circ + M_{\text{mineral}} s_{\text{mineral}}^\circ + M_{\text{SOC}} s_{\text{SOC}}^\circ + M_{\text{bio}} s_{\text{bio}}^\circ \quad (31)$$

Guaranteeing positive definite entropy and zero entropy destruction at initialization ensures that for any non-equilibrium state transition operator $\mathcal{T}$:

$$\Delta S_{\text{univ}} = \Delta S_{\text{cell}} + \Delta S_{\text{surroundings}} \geq 0 \quad (32)$$

5 Monadic Category Implementation

In functional programming abstractions of physical processes, physical state transitions can be formalized through monads [3]. We define the spatial monad $\mathcal{M}(S) = \text{SpatialMonad}\langle S \rangle$.

The factory function `createDefaultH3CellThermodynamicState` acts as the identity unit morphism η :

$$\eta : \text{H3Index} \longrightarrow \mathcal{M}(\text{IH3CellThermodynamicState}) \quad (33)$$

```
export function createDefaultH3CellThermodynamicState(
  h3Index: string,
  overrides?: Partial<IH3CellThermodynamicState>
): IH3CellThermodynamicState {
  const resolution = getH3Resolution(h3Index);
  const areaM2 = getH3CellAreaM2(h3Index);
  const T0 = STP_CONSTANTS.T_STANDARD; // 288.15 K
  const P0 = STP_CONSTANTS.P_STANDARD; // 101325.0 Pa

  // Total Column Mass Hydrostatically Derived
  const massAtmTotal = areaM2 * (P0 / STP_CONSTANTS.STANDARD_GRAVITY);
  const totalAtmosphericMoles = massAtmTotal / STP_CONSTANTS.MOLAR_MASS_WET_AIR
    ;

  const atmosphere: IThermodynamicAtmosphereStock = Object.freeze({
    nitrogenMoles: totalAtmosphericMoles * STP_CONSTANTS.MOLE_FRACTION_N2,
    oxygenMoles: totalAtmosphericMoles * STP_CONSTANTS.MOLE_FRACTION_O2,
    co2Moles: totalAtmosphericMoles * STP_CONSTANTS.MOLE_FRACTION_CO2,
    waterVaporMoles: totalAtmosphericMoles * STP_CONSTANTS.MOLE_FRACTION_H2O,
    surfacePressurePa: P0,
    ...overrides?.atmosphere
  });
  // Hydrosphere, Lithosphere, Biosphere instantiation
  return Object.freeze({ atmosphere, ... });
}
```

Figure 1: TypeScript implementation snippet for baseline state factory.

Monadic bind operations (`flatMap`) over these states preserve structural immutability through `Object.freeze`, ensuring that parallel thread execution and hexagonal neighbor diffusions cannot induce data races or conservation violations.

6 Verification and Invariant Analysis

The factory was validated across several numerical test suites:

1. **Hydrostatic Surface Pressure Closure:** The computed sum $\sum_i n_i M_i$ evaluated against $\frac{P_0 A}{g_0}$ yielded a relative discrepancy $\epsilon < 10^{-12}$, confirming exact conservation of molar column masses.
2. **Positive-Definite Mass & Temperature:** For all H3 cells across resolutions $r \in [0, 15]$, mass arrays satisfy $M_k > 0$, and temperature strictly satisfies $T > 0 \text{ K}$.
3. **Immutability Integrity:** Mutation attempts on nested stock objects throw strict runtime `TypeError` exceptions under ECMAScript 2022 strict mode.

7 Conclusion and Future Outlook

The baseline STP thermodynamic state tensor factory established in Sprint 044 provides a rigorous, physically grounded initialization primitive for hexagonal discrete global grid simulations. By joining geodesic geometry, hydrostatic atmospheric physics, psychrometric vapor balances, and pedospheric-biospheric stoichiometric inventories within a functional monadic container, the Web of Life architecture guarantees First and Second Law fidelity from simulation inception. Subsequent sprints will deploy this unit primitive in spatial diffusion monads, trophic predator-prey flux operators, and multi-scale climate-biosphere feedback loops.

Repository Availability

The implementation and verification suites are fully open-source and reproducible at:
<https://github.com/pascalranoroarijaona/WebOfLife>.

References

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